

Summary

Computer Science student skilled in building **scalable full-stack mobile** and **backend systems**. Experienced in designing and implementing **RESTful backend systems** using **Java (Spring Boot)** and **Python**, supported by **strong OOP principles** and **Test-Driven Development** practices. Able to work across the stack, including **responsive web interfaces**, **relational databases**, and **cloud deployment**. Strong collaborator with a team-first mindset and precise attention to logic for correct implementation

Education

Honours Bachelor of Science in Computer Science, York University, Toronto Expected Oct 2027
Relevant Coursework: Algorithms, Data Structures, Database Systems, Advanced OOP, User Interfaces

Relevant Projects

Bank Mobile App

- Built a **RESTful Java full-stack banking system** using a **layered architecture** (controller, service, repository) via **Spring Boot & PostgreSQL**, for **maintainability** through **modular design**
- Applied **TDD, JUnit, Mockito**, and **Spring Test (MockMvc)**, covering **unit, controller** and **service integration-level tests** for **code reliability** & **efficient debugging** through **early defect detection**
- Designed **relational schema & integrated PostgreSQL** via **Spring Data JPA** for **persistent storage**
- Designed and implemented **5 REST API endpoints** by analyzing user interface flow (account creation, deposit, withdraw, transfer, statement generation)
- Integrated **centralized input validation** in a **core service layer** for **reliable transaction processing**
- Implemented **client UI** in **Android Studio**, integrating **REST APIs** for real-time account operations
- **Dockerized backend app** & deployed to **AWS Lightsail**, for **cloud-based access on mobile app**

Website Portfolio (includes demo videos of projects)

- Utilized **Bootstrap Grid System, HTML** to create **5 responsive pages** via **breakpoints utilities** to support **different screen sizes**
- Developed **reusable UI components** using **custom elements** to ensure **maintainability**
- Applied **Javascript** and **CSS** to implement **dynamic UI behavior**, improving **content visibility** (e.g. navigation bar transitions based on scroll events)
- Implemented contact page with **input validation** & user-friendly feedback, improving **error recovery**

Driving Simulation Game

- Built a **Python-based** driving simulation with **OOP** to model user progression on a virtual road
- Utilized **PyGame API** to develop application logic and UI elements (e.g. **7 vehicles**, dynamic progress bar)
- Designed **dynamic generation of bot vehicles & rule-based movement logic** with **attention to detail** to produce intentional collision scenarios, improving **user engagement**
- Developed **4 termination conditions** to handle **state transitions** involving **bounding box collision detection** with user & bots, screen boundary violations, and finish-line completion logic

Technical Skills

Languages: Java, Python, SQL, HTML, CSS, Javascript, Bash, C

Backend & Frameworks: Spring Boot, Spring Data JPA, REST APIs, JUnit 5, MockMvc, Mockito

Other: PostgreSQL, Docker, AWS Lightsail, Android Studio, Git/GitHub, Bootstrap, Figma

Volunteering

Orientation Logistics Volunteer

August 2019

University of Toronto

- Assembled **8000+ student orientation kits** with student supplies in a **fast-paced, deadline-driven environment** through **constant communication** and **teamwork** with **100+ volunteers**
- Collaborated with **volunteers** to manage **5000+ by-products** from packing to maintain an **organized workspace**